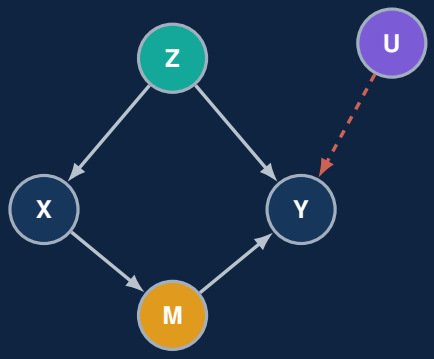


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 13

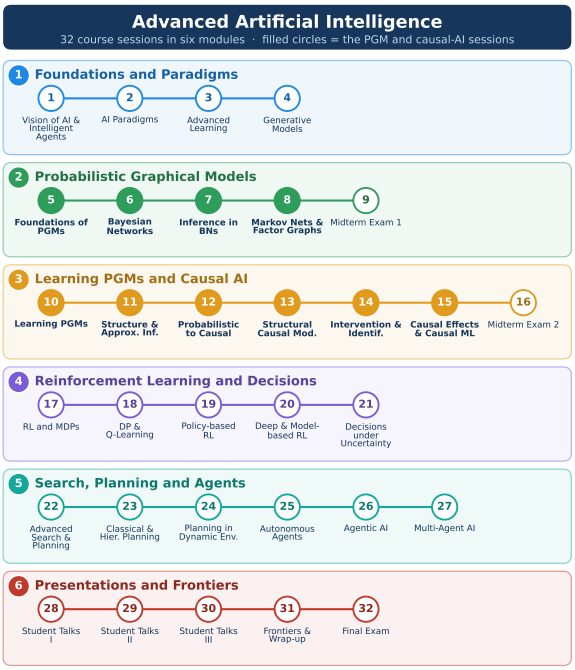
Structural Causal Models

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 13 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s13_examples.py` for the numbers, `python/s13_figures.py` for the figures), and `questions/s13_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 13.

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SESSION 13 ADVANCED ARTIFICIAL INTELLIGENCE

Structural Causal Models

Structural equations, causal DAGs, the causal Markov condition, d-separation and counterfactuals

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Structural Equations
Mechanisms, assignments, push-forward; structural vs statistical equations.

PART II
Causal DAGs & d-Separation
Causal Markov condition, testable implications, faithfulness, latent confounders.

PART III
Interventions & Counterfactuals
Graph surgery, truncated factorization, abduction–action–prediction, PN/PS.

PART IV
Mathematical Statistics
Linear SEMs, omitted-variable bias, regression on parents, linear counterfactuals.

PART V
AI Connections
World models, agent what-ifs, COMA credit assignment, counterfactual XAI.

LEARNING OBJECTIVES

- Define a structural causal model $M = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}))$ and explain why $:=$ is an *assignment*, not an equation.
- State and prove the **causal Markov condition**: a Markovian SCM induces a distribution that factorizes over its causal DAG.
- Read **testable implications** off a causal DAG by d-separation, and recognise faithfulness, minimality and latent confounding.
- Compute interventional distributions by graph surgery and the truncated factorization, and counterfactuals by abduction, action and prediction.
- Derive the covariance of a linear SEM, $\Sigma = (I-B)^{-1}\Omega(I-B)^{-\top}$, and separate regression coefficients from structural coefficients.
- Use SCM reasoning in AI: world models in model-based RL, "what-if" simulation in LLM agents, counterfactual credit assignment (COMA) in multi-agent RL.

Where we are. Sessions 5–11 (course sessions 5–11) built the *probabilistic* machinery: conditional independence, Bayesian and Markov networks, exact and approximate inference, parameter and structure learning. Session 12 made the conceptual pivot: correlation is not causation, $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$, and confounding and selection bias explain the gap. Today we build the mathematical object that makes $\mathrm{do}(\cdot)$ and "what would have happened" well defined: the **structural causal model**. Session 14 will then ask when causal quantities can be *identified* from observational data,

and Session 15 will estimate them with machine learning. Beyond the notes, SCMs reappear in model-based RL, planning, agentic and multi-agent AI (course sessions 20–27, 31).

OPENING MOTIVATION

Session 12 ended on a gap: $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$ in general, and nothing in our toolkit so far can compute the right-hand side, because a DAG with CPTs only ever specified a *factorization* of P — it never said what happens if a piece of that factorization is overridden rather than observed. Today we fix this by strengthening what an arrow means. Instead of "X_i's distribution depends on its parents," a structural causal model (SCM) says "X_i is *computed from* its parents by a fixed mechanism" — a claim strong enough to survive having one mechanism swapped out, which is exactly what an intervention does.

PART I

Structural Equations

Mechanisms, not merely factorizations

PART I · STRUCTURAL EQUATIONS

13.1 Structural Causal Models

Definition 13.1 (Structural causal model)**.** A structural causal model $M = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}))$ consists of:

- *exogenous* variables $\mathbf{U}$ — background factors external to the model, with a specified joint distribution $P(\mathbf{U})$;
- *endogenous* variables $\mathbf{V} = \{V_1, \ldots, V_n\}$ — the variables of interest, determined within the model;
- a set of *structural equations* $\mathcal{F} = \{f_1, \ldots, f_n\}$, one per endogenous variable, of the form

$$V_i := f_i\big(\mathrm{pa}(V_i), U_i\big),$$

where $\mathrm{pa}(V_i) \subseteq \mathbf{V} \setminus \{V_i\}$ and $U_i \subseteq \mathbf{U}$ are the direct causes of V_i.

The symbol $:=$ (assignment, not equality) is deliberate: $V_i := f_i(\cdots)$ says V_i is *computed* by this mechanism, not merely that this equation happens to hold. This asymmetry — present in the notation but invisible in an ordinary equals sign — is precisely what will let us "delete" one equation while keeping the rest, which an algebraic equality could never justify (you cannot selectively erase one equation from a system of equalities without changing what the whole system claims).

Every SCM induces an ordinary graph G: draw an edge $V_j \to V_i$ whenever $V_j \in \mathrm{pa}(V_i)$. We call M *recursive* (acyclic) if G is a DAG, and *Markovian* if in addition the noises $U_1, \ldots, U_n$ are mutually independent. Unless stated otherwise all SCMs in these notes are recursive. And crucially:

Proposition 13.1 (Consistency with Bayesian networks (Session 6)). *The distribution $P(\mathbf{v})$ induced by a Markovian SCM (obtained by pushing $P(\mathbf{U})$ forward through $\mathcal{F}$) factorizes exactly as $\prod_i P(v_i \mid \mathrm{pa}(v_i))$ over the induced DAG G — the same Bayesian network factorization from Session 6.*

So nothing from Sessions 5–11 is discarded: an SCM is a *refinement* of a Bayesian network, adding mechanism on top of factorization, and every observational computation (VE, MLE, EM, sampling) still applies unchanged to the induced $P(\mathbf{v})$. What's new is that the SCM also licenses questions the factorization alone could not answer. Proposition 13.1 is proved in full as the causal Markov condition, Theorem 13.1 below.

Worked Example 13.1 (A linear SCM for wages). Let A (ability), X (years of education), Y (income) be linked by

$$A := U_A, \qquad X := 2A + U_X, \qquad Y := 3X + 5A + U_Y,$$

with U_A, U_X, U_Y mutually independent standard normal noises. For later use we name the three structural coefficients $a = 2$ (edge $A \to X$), $b = 3$ (edge $X \to Y$) and $c = 5$ (edge $A \to Y$). This is exactly the confounding structure of Session 12: A is a common cause of X and Y (a fork), so observing X carries information about A, which in turn carries information about Y through the direct arrow $A \to Y$ — inflating any naive association between X and Y beyond the true structural coefficient 3.

PART I · STRUCTURAL EQUATIONS

13.2 Structural versus Statistical Equations

A regression equation and a structural equation can look identical on paper. The difference is what they promise under change.

Worked Example 13.2 (One mechanism, two regressions). Let $X := U_X$, $Y := 2X + U_Y$ with $U_X, U_Y \overset{\text{iid}}{\sim} N(0,1)$. Then (X, Y) is bivariate normal with zero means and

$$\mathrm{Var}(X) = 1, \qquad \mathrm{Var}(Y) = 2^2\mathrm{Var}(X) + \mathrm{Var}(U_Y) = 4 + 1 = 5,$$

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, 2X + U_Y) = 2\mathrm{Var}(X) = 2.$$

The best linear predictor of X from Y has slope $\mathrm{Cov}(X,Y)/\mathrm{Var}(Y) = 2/5 = 0.4$, i.e. $X = 0.4\,Y + E$, where $E = X - 0.4Y$ is uncorrelated with Y (hence, being jointly Gaussian with Y, independent of it) and $\mathrm{Var}(E) = \mathrm{Var}(X) - \mathrm{Cov}(X,Y)^2/\mathrm{Var}(Y) = 1 - 4/5 = 0.2$: a perfectly valid *statistical* equation. But it is not *structural*. Under $\mathrm{do}(Y = 5)$ the SCM replaces only Y's assignment; $X := U_X$ is untouched, so $\mathbb{E}[X \mid \mathrm{do}(Y = 5)] = \mathbb{E}[U_X] = 0$, whereas the regression "predicts" $\mathbb{E}[X \mid Y = 5] = 0.4 \cdot 5 = 2$. And solving $Y := 2X + U_Y$ for $X = (Y - U_Y)/2$ is algebraically valid but causally meaningless: it would claim that setting Y moves X.

PYTHON LAB · `s13_examples.py` · `ex_structural_vs_statistical()`

Simulation of the SCM with $n = 100\,000$ draws (`default_rng(11)`); the interventional mean uses fresh draws of $X := U_X$.

```
exact:  slope Cov/Var(Y) = 2/5 = 0.40, residual var = 0.20
simulated slope = 0.400, residual var = 0.199, corr(resid, Y) = -0.0000
E[X | do(Y=5)] (simulated) = -0.001  vs  E[X | Y=5] = 0.4*5 = 2.0
```

The property that distinguishes a structural equation from a regression equation has a name: **autonomy** or **modularity** — each mechanism f_i can be changed without changing the others. It is the assumption that makes interventions predictable, and it is what a world model in RL or a simulator inside a planning agent silently relies on.

Worked Example 13.3 (Two SCMs, one distribution, two causal answers)**.** Let M_1: $X := U_X$, $Y := 0.6X + U_Y$ with $U_X \sim N(0,1)$, $U_Y \sim N(0, 0.64)$; and M_2: $Y := V_Y$, $X := 0.6Y + V_X$ with $V_Y \sim N(0,1)$, $V_X \sim N(0, 0.64)$. Both induce a zero-mean bivariate normal law. Under M_1: $\mathrm{Var}(X) = 1$, $\mathrm{Var}(Y) = 0.6^2 \cdot 1 + 0.64 = 0.36 + 0.64 = 1$ and $\mathrm{Cov}(X, Y) = 0.6\mathrm{Var}(X) = 0.6$; under M_2, by the same computation with the roles swapped, $\mathrm{Var}(Y) = 1$, $\mathrm{Var}(X) = 0.36 + 0.64 = 1$ and $\mathrm{Cov}(X, Y) = 0.6$. A zero-mean Gaussian law is determined by its covariance matrix, so both models induce the same law (unit variances, correlation 0.6), and no amount of observational data can distinguish them. Yet under M_1 surgery keeps $Y := 0.6X + U_Y$, so $\mathbb{E}_{M_1}[Y \mid \mathrm{do}(X = 2)] = 0.6 \cdot 2 = 1.2$, while under M_2 the variable Y is a root whose assignment $Y := V_Y$ is untouched by $\mathrm{do}(X = 2)$, so $\mathbb{E}_{M_2}[Y \mid \mathrm{do}(X = 2)] = \mathbb{E}[V_Y] = 0$. The same construction (regress either variable on the other, Gaussian residual) works for *every* bivariate normal law. Observational equivalence does not imply interventional equivalence; this is the SCM version of Session 6's Markov equivalence, and the reason Session 14 needs identification theory.

PYTHON LAB · s13_examples.py · ex_markov_equivalent_gauss()

```
M1:  Var(Y) = 0.6^2*1 + 0.64 = 1.00, Cov(X,Y) = 0.6;  M2:  Var(X) = 1.00,
Cov = 0.6
Cov under M1 = Cov under M2 = [[1.  0.6] [0.6 1.  ]]
E[Y | do(X=2)]:   M1 -> 0.6*2 = 1.2,   M2 -> E[V_Y] = 0.0
```

PART II

Causal DAGs, the Causal Markov Condition and d-Separation

What a causal graph claims, and how the data can refute it

PART II · CAUSAL MARKOV

13.3 Causal DAGs and the Causal Markov Condition

Definition 13.2 (Causal DAG)**.** A DAG G over $\mathbf{V}$ is a *causal DAG* for a system if (i) every edge $V_j \to V_i$ means V_j is a *direct cause* of V_i relative to $\mathbf{V}$ (setting V_j by intervention can change V_i even when all other variables in $\mathbf{V}$ are held fixed), and (ii) every common cause of

two or more variables of $\mathbf{V}$ is itself in $\mathbf{V}$ (*causal sufficiency*). Equivalently, G is the graph of a Markovian SCM.

The *presence* of an edge is a weak claim (the effect may be tiny); the *absence* of an edge is a strong one (no direct effect) and, with causal sufficiency, generates the testable implications below.

Theorem 13.1 (Causal Markov condition)**.** *Let M be a Markovian SCM with causal DAG G. Then for every i,*

$$V_i \perp\!\!\!\perp \mathrm{nd}(V_i) \setminus \mathrm{pa}(V_i) \mid \mathrm{pa}(V_i) \qquad \textit{(local Markov property)},$$

and consequently the induced distribution factorizes as $P(\mathbf{v}) = \prod_{i=1}^{n} P(v_i \mid \mathrm{pa}(v_i))$, with $P(v_i \mid \mathrm{pa}_i) = P\big(f_i(\mathrm{pa}_i, U_i) = v_i\big)$.

Proof. Order the variables topologically. By induction along this order, each V_j is a (measurable) function of the noises $\{U_k : k \in \mathrm{an}(j) \cup \{j\}\}$ of itself and its ancestors. Fix i and let $N = \mathrm{nd}(V_i)$. If $V_j \in N$, then neither V_i nor any descendant of V_i is an ancestor of V_j (otherwise V_j would be a descendant of V_i). Hence $N = g(\mathbf{U}_S)$ for an index set S with $i \notin S$, and Markovianity gives $U_i \perp\!\!\!\perp N$. Since $\mathrm{pa}(V_i) \subseteq N$, for any measurable set B,

$$P(V_i \in B \mid N = n) = P\big(f_i(\mathrm{pa}_i(n), U_i) \in B \mid N = n\big) = P\big(f_i(\mathrm{pa}_i(n), U_i) \in B\big),$$

where the last step is the "freezing" lemma: if $U \perp\!\!\!\perp N$ then $P(h(N, U) \in B \mid N = n) = P(h(n, U) \in B)$, by Fubini's theorem on the product law $P_N \otimes P_U$. The right-hand side depends on n only through $\mathrm{pa}_i(n)$, which is the local Markov property. For the factorization, apply the chain rule along the topological order: the predecessors $V_{<i}$ are non-descendants of V_i, so $P(v_i \mid v_{<i}) = P(v_i \mid \mathrm{pa}_i)$. □

PART II · CAUSAL MARKOV

13.4 d-Separation in SCMs: Testable Implications

Because the induced distribution factorizes over G, everything we proved about Bayesian networks in Session 6 applies verbatim.

Theorem 13.2 (Global Markov property of SCMs)**.** *If M is a Markovian SCM with causal DAG G, then for disjoint sets $\mathbf{X}, \mathbf{Y}, \mathbf{Z} \subseteq \mathbf{V}$,*

$$\mathbf{X} \perp\!\!\!\perp_d \mathbf{Y} \mid \mathbf{Z} \text{ in } G \quad \Longrightarrow \quad \mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z} \text{ in } P_M.$$

Proof sketch. P_M factorizes over G (Theorem 13.1), and by the soundness theorem of d-separation (Session 6; Verma and Pearl) every such distribution satisfies all d-separations of G. □

Every d-separation in the causal DAG is therefore a **testable implication** of the causal model: a constraint on observational data that the model must satisfy, whatever the mechanisms f_i are. If the data violate it, the causal model (or the Markov assumption) is wrong.

13.4.1 Faithfulness and minimality

Theorem 13.2 is one-directional: d-separation implies independence, but a distribution may contain *extra* independences.

Definition 13.3 (Faithfulness and minimality)**.** P is *faithful* to G if every CI in P is implied by d-separation in G, i.e. $\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z} \Rightarrow \mathbf{X} \perp\!\!\!\perp_d \mathbf{Y} \mid \mathbf{Z}$. P satisfies *causal minimality* with respect to G if it is Markov to G but not to any proper subgraph of G (every edge is "used").

Worked Example 13.4 (Cancelling paths)**.** Let $X := U_X$, $M := X + U_M$, $Y := bX + M + U_Y$ with unit-variance independent Gaussian noises. Then $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, bX + M + U_Y) = b\mathrm{Var}(X) + \mathrm{Cov}(X, M) = b \cdot 1 + 1 = b + 1$, so at $b = -1$ the direct and mediated effects cancel exactly: $X \perp\!\!\!\perp Y$ although $X \to Y$ is an edge, and the total effect $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(x)] = b + 1 = 0$. The distribution is Markov and minimal but *not faithful*. The violation is visible in other statistics. With $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M) = 1 + 1 = 2$ and $\mathrm{Cov}(X, M) = 1$, Gaussian conditioning gives $\mathrm{Var}(X \mid M) = 1 - 1^2/2 = \frac{1}{2}$; since $Y = bX + M + U_Y$ and M is fixed, $\mathrm{Cov}(X, Y \mid M) = b\mathrm{Var}(X \mid M) = -\frac{1}{2}$ and $\mathrm{Var}(Y \mid M) = b^2\mathrm{Var}(X \mid M) + \mathrm{Var}(U_Y) = \frac{1}{2} + 1 = \frac{3}{2}$, so

$$\rho_{XY \cdot M} = \frac{-1/2}{\sqrt{(1/2)(3/2)}} = \frac{-0.5}{\sqrt{0.75}} = -0.577 \neq 0.$$

The pattern "$X \perp\!\!\!\perp Y$ but $X \not\perp\!\!\!\perp Y \mid M$" is exactly what a *faithful collider* $X \to M \leftarrow Y$ would produce, so a constraint-based learner that assumes faithfulness (such as PC) would output that wrong graph.

PYTHON LAB · `s13_examples.py` · `ex_faithfulness()`

```
Cov(X,Y) exact = b + 1 = -1 + 1 = 0; simulated (n=100000, seed=5) corr(X,Y)
= +0.0020
pcorr(X,Y|M) simulated = -0.5786;  exact via conditional covariances:
-0.5774
```

For linear-Gaussian SCMs with coefficients drawn from a density, unfaithful parameters have Lebesgue measure zero (Spirtes, Glymour and Scheines 2000): faithfulness is a mild *generic* assumption — the one the PC algorithm of Session 11 needs — but near-cancellations (feedback, agents that *compensate* for disturbances) make it fragile in finite samples.

PART II · CAUSAL MARKOV

13.5 Semi-Markovian Models and Bidirected Edges

If two noises are dependent — typically because an unmeasured variable L affects both V_i and V_j — the SCM is *semi-Markovian*. We then drop L from the picture and draw a dashed bidirected edge $V_i \leftrightarrow V_j$, obtaining an *acyclic directed mixed graph* (ADMG). The visible DAG part no longer satisfies Theorem 13.1, but d-separation extends to ADMGs (m-separation), treating $V_i \leftrightarrow V_j$ as $V_i \leftarrow L \rightarrow V_j$. The wage SCM with A unmeasured is the "bow-arc" ADMG $X \to Y$, $X \leftrightarrow Y$, which Session 14 shows is *not* identifiable; bidirected edges partition $\mathbf{V}$ into *c-components*, the building blocks of the ID algorithm.

PART III

Interventions and Counterfactuals

From seeing, to doing, to imagining

PART III · INTERVENTIONS

13.6 Intervention as Graph Surgery

Definition 13.4 (Intervention)**.** The intervention $\mathrm{do}(X = x)$ on SCM M produces a new (*mutilated*, or *submodel*) SCM M_x by replacing the structural equation for X with the constant assignment $X := x$, deleting X's dependence on $\mathrm{pa}(X)$ and U_X, and leaving every other equation in $\mathcal{F}$ unchanged. The interventional distribution $P(\mathbf{y} \mid \mathrm{do}(x))$ is the distribution of $\mathbf{Y}$ induced by M_x.

Graphically (Fig. 13.1), this deletes every incoming edge to X while keeping all outgoing edges intact — X still causes whatever it caused before, it is simply no longer caused by anything itself.

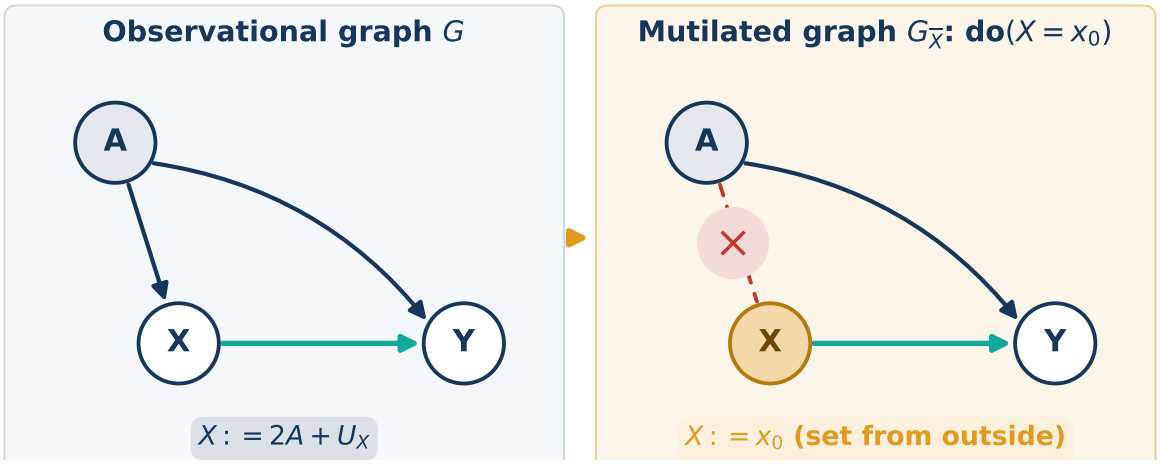


Figure 13.1. Graph surgery: $\mathrm{do}(X = x_0)$ deletes X's incoming edges (here, from confounder A) while keeping its outgoing edges to Y.

Theorem 13.3 (Truncated factorization)**.** *Let M be a Markovian SCM over $\mathbf{V} = \{V_1, \ldots, V_n\}$ and let $X = V_k$ be one of its variables. For every configuration $(v_1, \ldots, v_n)$,*

$$P(v_1, \ldots, v_n \mid \mathrm{do}(X = x)) = \begin{cases} \prod_{i \neq k} P\big(v_i \mid \mathrm{pa}(v_i)\big) & \text{if } v_k = x, \\ 0 & \text{if } v_k \neq x, \end{cases}$$

i.e. the configuration must agree with the intervention, and any parent value $\mathrm{pa}(v_i)$ that involves X is evaluated at $X = x$. In words: drop the factor $P(x \mid \mathrm{pa}(x))$ from the Bayesian-network factorization of Theorem 13.1 (Session 6) and fix $X = x$ everywhere else.

Proof. M_x is again a Markovian SCM (its noises are a subset of the original independent noises), with graph $G_{\overline{X}}$. By Theorem 13.1, $P_{M_x}(\mathbf{v}) = \prod_i P_{M_x}(v_i \mid \mathrm{pa}_i)$. For $V_i \neq X$ the mechanism and noise are unchanged, so $P_{M_x}(v_i \mid \mathrm{pa}_i) = P(f_i(\mathrm{pa}_i, U_i) = v_i) = P(v_i \mid \mathrm{pa}_i)$; the factor of X is the point mass $\mathbb{1}(v_X = x)$. (The kernels $P(v_i \mid \mathrm{pa}_i)$ are defined by the mechanisms for every parent value, so no positivity assumption is needed in the model; *estimating* them from data does require the relevant parent configurations to occur.) □

This is the single most important formula of this session, and it is the bridge connecting the entire directed-graphical-model machinery of Sessions 6–7 directly to causal queries: computing an interventional distribution is *exactly* an ordinary Bayesian-network computation (Session 7's variable elimination applies immediately) on the *mutilated* graph, using the very same CPTs as the original network for every equation except X's. No new algorithm is needed — only a different graph.

INTUITION

Contrast conditioning with intervening using our confounding example. Conditioning on $X = x$ (Session 12's $P(y \mid x)$) restricts attention to a subpopulation — everyone in that subpopulation still got their X value through the ordinary mechanism $X := 2A + U_X$, so if X is high partly because A is high, that information about A leaks into what we know about Y. Intervening, $\mathrm{do}(X = x)$, severs this: every unit is forced to $X = x$ regardless of its A, so the leak is cut at the source, exactly as Theorem 13.3 formalizes by simply deleting X's factor.

Summing the truncated factorization over all variables other than Y and $\mathbf{Z} = \mathrm{pa}(X)$ gives the **adjustment formula**

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z})\, P(\mathbf{z}) \qquad (\mathbf{Z} = \mathrm{pa}(X),\ Y \notin \mathrm{pa}(X)).$$

Derivation (assuming $P(x \mid \mathbf{z}) > 0$): for every configuration $\mathbf{v}$ with $v_k = x$, Theorem 13.3 says $P(\mathbf{v} \mid \mathrm{do}(x)) = P(\mathbf{v})/P(x \mid \mathbf{z})$, because the full product $P(\mathbf{v})$ differs from the truncated one only by the factor $P(x \mid \mathbf{z})$. Summing over all variables except Y and $\mathbf{Z}$ (the denominator depends only on $\mathbf{z}$) gives $P(y, \mathbf{z} \mid \mathrm{do}(x)) = P(y, x, \mathbf{z})/P(x \mid \mathbf{z}) = P(y \mid x, \mathbf{z})P(\mathbf{z})$; finally sum over $\mathbf{z}$. For the wage SCM this reads $P(y \mid \mathrm{do}(x)) = \int p(y \mid x, a)\, p(a)\, da$, and for the discrete examples below it is a two-term sum over the confounder. A *soft* intervention replaces f_X by a new mechanism such as a policy $\pi(x \mid \mathbf{z})$, which simply replaces the factor $P(x \mid \mathrm{pa}_x)$ — the way every RL policy acts on its environment.

PART III · INTERVENTIONS

13.7 Counterfactuals

Interventions ask "what would happen to a population if we set $X = x$?" Counterfactuals ask a sharper, unit-specific question: "for this specific unit, who we observed with $X = x$ and $Y = y$, what would Y have been had X been x' instead?" SCMs answer this via a three-step procedure:

1. **Abduction**: given the observed evidence e, replace the prior $P(\mathbf{U})$ by the posterior $P(\mathbf{U} \mid e)$. If every endogenous variable of the unit is observed *and* each mechanism is invertible in its own noise (e.g. additive noise, $V_i := g_i(\mathrm{pa}_i) + U_i$, so $U_i = V_i - g_i(\mathrm{pa}_i)$), the posterior is a point mass and $\mathbf{U}$ is pinned down exactly; otherwise it is a genuine distribution.

2. **Action**: perform $\mathrm{do}(X = x')$ (graph surgery, Definition 13.4) on the model.
3. **Prediction**: compute Y in this modified model, using the *same* (abduced) value of $\mathbf{U}$ as this specific unit.

Definition 13.5 (Counterfactual)**.** For a unit $\mathbf{u}$, the *potential response* $Y_x(\mathbf{u})$ is the value of Y in the submodel M_x evaluated at $\mathbf{U} = \mathbf{u}$. For evidence e (an observed event about the endogenous variables), the counterfactual distribution is $P(Y_{x'} = y' \mid e) = \sum_{\mathbf{u}} \mathbb{1}\{Y_{x'}(\mathbf{u}) = y'\}\, P(\mathbf{u} \mid e)$ (an integral against $P(\mathbf{u} \mid e)$ for continuous noise). Two immediate consequences (Pearl 2009, ch. 7) are the *consistency* rule, $X(\mathbf{u}) = x \Rightarrow Y_x(\mathbf{u}) = Y(\mathbf{u})$ (setting X to the value it would have taken anyway changes nothing), and the *composition* rule, $W_x(\mathbf{u}) = w \Rightarrow Y_{x,w}(\mathbf{u}) = Y_x(\mathbf{u})$ (additionally setting W to the value it takes under $\mathrm{do}(x)$ changes nothing).

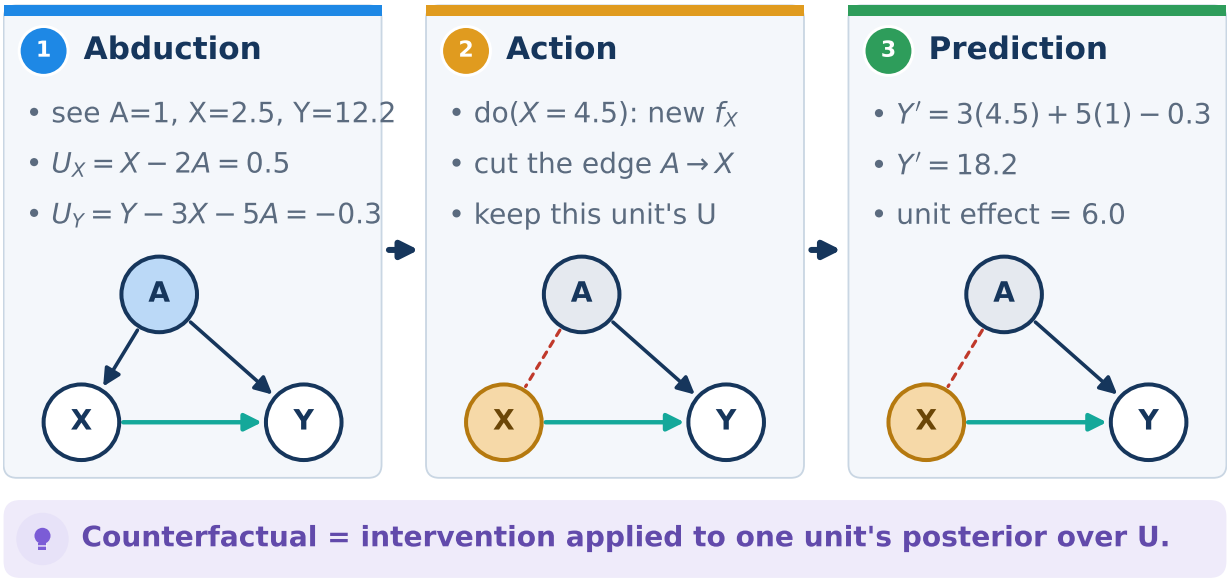


Figure 13.2. The three-step counterfactual computation on the wage SCM, for the individual of Worked Example 13.6.

Remark 13.1 (Bridge to potential outcomes). Students arriving from the potential-outcomes (Neyman–Rubin) tradition will recognize $Y_x(u)$ — "the value Y would take for unit u if X were set to x" — as exactly the SCM counterfactual just defined, usually written $Y(x)$ or Y_x. The two frameworks are provably equivalent in what they can express; SCMs simply make the generative mechanism explicit, which is convenient for reading identifiability directly off a graph (Session 14), while potential outcomes are often more convenient for stating assumptions directly in terms of estimands (Session 15). We will use both notations as convenient from here on.

13.7.1 Probabilities of causation

For binary X, Y, write x for the presence of the cause ($X = 1$), x' for its absence ($X = 0$), and y for the event $Y = 1$. Three counterfactual quantities capture "did X cause Y?" (Pearl 2009, ch. 9):

$$\mathrm{PN} = P(Y_{x'} = 0 \mid X = x, Y = 1), \quad \mathrm{PS} = P(Y_x = 1 \mid X = x', Y = 0), \quad \mathrm{PNS} = P(Y_x = 1, Y_{x'} = 0).$$

PN (probability of necessity) is the legal "but-for" question: among cases with the cause and the effect, how often would the effect have been absent without the cause? PS (probability of

sufficiency) asks the converse: among cases with neither cause nor effect, how often would the cause have produced the effect? PNS (probability of necessity and sufficiency) is the population share of units for which the effect occurs with the cause and not without it.

Proposition 13.2 (Bounds and monotone identification)**.** *Let* $p_1 = P(y \mid \mathrm{do}(x))$ *and* $p_0 = P(y \mid \mathrm{do}(x'))$*. Then* $\max(0, p_1 - p_0) \le \mathrm{PNS} \le \min(p_1, 1 - p_0)$*. If* Y *is* monotone *in* X *($Y_{x'} = 1 \Rightarrow Y_x = 1$ for every unit), then* $\mathrm{PNS} = p_1 - p_0$*; if moreover* X *is exogenous* $(Y_x, Y_{x'} \perp\!\!\!\perp X)$*, then* $\mathrm{PN} = (p_1 - p_0)/p_1$ *and* $\mathrm{PS} = (p_1 - p_0)/(1 - p_0)$*.*

Proof. Each unit has one of four *response types* $(Y_x, Y_{x'})$: always $(1,1)$, helped $(1,0)$, hurt $(0,1)$, never $(0,0)$, with probabilities a, h, k, n. Then $p_1 = a + h$, $p_0 = a + k$ and $\mathrm{PNS} = h$. From $h = p_1 - p_0 + k \ge p_1 - p_0$ and $h \ge 0$ we get the lower bound; from $h = p_1 - a \le p_1$ and $h \le 1 - a - k = 1 - p_0$ the upper bound. Monotonicity means $k = 0$, so $h = p_1 - p_0$. Under exogeneity the type distribution is the same in the $X = x$ and $X = x'$ groups; units with $X = x, Y = 1$ are of type always or helped (mass p_1), so $\mathrm{PN} = h/p_1$; units with $X = x', Y = 0$ are helped or never (mass $1 - p_0$), so $\mathrm{PS} = h/(1 - p_0)$. □

PART III · INTERVENTIONS

13.8 Which Way Does the Arrow Point? Additive-Noise Models

Worked Example 13.3 showed that for linear-Gaussian SCMs the direction of an edge is not identifiable from P. Surprisingly, Gaussianity is the *only* obstacle in the linear case. In an **additive-noise model** $Y := f(X) + U_Y$ with $U_Y \perp\!\!\!\perp X$, one fits regressions in both directions and checks which residual is independent of its regressor; for generic nonlinear f or non-Gaussian noise, only the causal direction admits an independent residual (Hoyer et al. 2009; Peters, Janzing and Schölkopf 2017). In the linear case this works whenever the noises are non-Gaussian (LiNGAM; Shimizu et al. 2006).

PART IV

Mathematical Statistics of SCMs

Linear SEMs, regression and linear counterfactuals

PART IV · MATH STATISTICS

13.9 Linear SEMs and Regression

A *linear structural equation model* (SEM) is an SCM $V_i := \sum_j b_{ij} V_j + U_i$ with $b_{ij} \neq 0$ only if $V_j \to V_i$, independent noises with $\mathrm{Var}(U_i) = \omega_i$. In matrix form $\mathbf{V} = B\mathbf{V} + \mathbf{U}$ with $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_n)$.

Theorem 13.4 (Covariance of a linear SEM)**.** *If* G *is acyclic,* $I - B$ *is invertible,* $\mathbf{V} = (I - B)^{-1}\mathbf{U}$*, and*

$$\Sigma = \mathrm{Cov}(\mathbf{V}) = (I - B)^{-1}\, \Omega\, (I - B)^{-\top}.$$

Proof. In a topological order B is strictly lower triangular, so $I - B$ is unit lower triangular with $\det(I - B) = 1$; hence it is invertible and $\mathbf{V} = B\mathbf{V} + \mathbf{U}$ has the unique solution $\mathbf{V} = (I - B)^{-1}\mathbf{U}$. Then $\mathrm{Cov}(\mathbf{V}) = (I - B)^{-1}\mathrm{Cov}(\mathbf{U})(I - B)^{-\top} = (I - B)^{-1}\Omega(I - B)^{-\top}$. □

Proposition 13.3 (Omitted-variable bias)**.** *Let $Y := \mathbf{b}^\top\mathbf{X} + \mathbf{c}^\top\mathbf{A} + U_Y$ with $U_Y \perp\!\!\!\perp (\mathbf{X}, \mathbf{A})$ and Σ_{XX} invertible. The population least-squares coefficient of Y on $\mathbf{X}$ alone is*

$$\boldsymbol{\beta} = \Sigma_{XX}^{-1}\Sigma_{XY} = \mathbf{b} + \Sigma_{XX}^{-1}\Sigma_{XA}\,\mathbf{c}.$$

In the scalar case with $X := aA + U_X$, $\mathrm{Var}(A) = \sigma_A^2$, $\mathrm{Var}(U_X) = \sigma_X^2$: $\beta = b + ac\sigma_A^2/(a^2\sigma_A^2 + \sigma_X^2)$.

Proof. $\Sigma_{XY} = \mathrm{Cov}(\mathbf{X}, \mathbf{b}^\top\mathbf{X} + \mathbf{c}^\top\mathbf{A} + U_Y) = \Sigma_{XX}\mathbf{b} + \Sigma_{XA}\mathbf{c} + \mathbf{0}$; multiply by Σ_{XX}^{-1}. In the scalar case $\Sigma_{XA} = a\sigma_A^2$ and $\Sigma_{XX} = a^2\sigma_A^2 + \sigma_X^2$. □

The bias term $\Sigma_{XX}^{-1}\Sigma_{XA}\mathbf{c}$ is the coefficient of the auxiliary regression of $\mathbf{A}$ on $\mathbf{X}$ times the effect of $\mathbf{A}$ on Y: it vanishes iff the omitted variables are uncorrelated with $\mathbf{X}$ or irrelevant for Y — the algebraic face of "confounding needs both arrows of the fork".

Proposition 13.4 (Regression on the parents recovers the mechanism)**.** *In a Markovian linear SEM with $\Omega \succ 0$, the population regression of V_i on $\mathrm{pa}(V_i)$ has coefficients $(b_{ij})_{j\in\mathrm{pa}(i)}$ and residual variance ω_i.*

Proof. $V_i = \mathbf{b}_i^\top\mathrm{pa}_i + U_i$ with $U_i \perp\!\!\!\perp \mathrm{pa}_i$ by Theorem 13.1 (parents are non-descendants). Hence $\mathrm{Cov}(\mathrm{pa}_i, V_i - \mathbf{b}_i^\top\mathrm{pa}_i) = \mathrm{Cov}(\mathrm{pa}_i, U_i) = \mathbf{0}$, which are the normal equations; they have a unique solution because $\Sigma_{\mathrm{pa}_i\mathrm{pa}_i}$ is a principal submatrix of $\Sigma \succ 0$ (Theorem 13.4 with $\Omega \succ 0$). The residual is U_i, with variance ω_i. □

MATHEMATICAL STATISTICS · Regression coefficients versus structural coefficients

A regression coefficient belongs to P *and the chosen regressor set*; a structural coefficient belongs to a mechanism. They coincide for the parent set (Proposition 13.4). The coefficient of X is the *total* effect when the other regressors block every back-door path and contain no descendant of X (here $\{A\}$: 0.72), and the *direct* effect when they also block mediated paths ($\{A, M\}$: 0.3); colliders or descendants of Y bias it again. Choosing regressors is a causal decision — Session 14's back-door criterion.

Theorem 13.5 (Counterfactuals without effect heterogeneity)**.** *Suppose $Y_x(\mathbf{u}) = \tau x + g(\mathbf{u})$ for all $x, \mathbf{u}$ (constant effect τ, e.g. any linear SEM with additive noise). Then for any evidence e that includes $X = x$, the counterfactual law of $Y_{x'}$ is the factual law of Y given e shifted by $\tau(x' - x)$: in particular $\mathbb{E}[Y_{x'} \mid e] = \mathbb{E}[Y \mid e] + \tau(x' - x)$.*

Proof. For every $\mathbf{u}$ consistent with e, consistency gives $Y(\mathbf{u}) = Y_x(\mathbf{u})$, so $Y_{x'}(\mathbf{u}) = \tau x' + g(\mathbf{u}) = Y(\mathbf{u}) + \tau(x' - x)$. Push the posterior $P(\mathbf{u} \mid e)$ forward through both sides. □

Theorem 13.5 explains why abduction looked trivial in the linear wage example: the unit-level effect does not depend on $\mathbf{u}$. With heterogeneity it does, and abduction matters.

PART IV · MATH STATISTICS

13.10 Worked Examples

Worked Example 13.5 (The wage SCM: observational vs. interventional slope)**.** Using Example 13.1, compute (i) the slope of the observational regression of Y on X, and (ii) the true causal effect $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(X = x)]$.
Solution. Substituting $X = 2A + U_X$ into $Y = 3X + 5A + U_Y$ gives $Y = 6A + 3U_X + 5A + U_Y = 11A + 3U_X + U_Y$. With unit-variance independent noises, $\mathrm{Cov}(X, Y) = \mathrm{Cov}(2A + U_X,\ 11A + 3U_X + U_Y) = 22\,\mathrm{Var}(A) + 3\,\mathrm{Var}(U_X) = 22 + 3 = 25$, and $\mathrm{Var}(X) = 4\,\mathrm{Var}(A) + \mathrm{Var}(U_X) = 4 + 1 = 5$, giving observational slope $25/5 = 5$. For the interventional quantity, Theorem 13.3 says: drop X's equation, fix $X = x$, keep $A := U_A$ and $Y := 3X + 5A + U_Y$ unchanged; then $\mathbb{E}[Y \mid \mathrm{do}(x)] = 3x + 5\mathbb{E}[A] + \mathbb{E}[U_Y] = 3x$ (since $\mathbb{E}[A] = \mathbb{E}[U_Y] = 0$), so $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(x)] = 3$ exactly — matching the structural coefficient by construction, and matching Session 12's warning precisely: the naive association (5) overstates the true causal effect (3) by exactly the confounding contribution $ac/(a^2 + 1) = 2 \cdot 5/(4 + 1) = 2$, with $a = 2$, $c = 5$ from Example 13.1 (Proposition 13.3). A simulation confirms both values (`simulate_wage()`, `interventional_curve()`): an ordinary least-squares regression of Y on X for $n = 2000$ draws of the SCM (`default_rng(123)`) has slope 5.03, while directly simulating the intervention — at each of 30 equally spaced values x spanning the observed range of X, set $X := x$ and average $Y = 3x + 5A + U_Y$ over 5000 fresh draws of (A, U_Y) (`default_rng(7)`) — and fitting a line through the 30 averages gives slope 3.00.

PYTHON LAB · s13_examples.py · ex_wage_slopes()

```
exact:  Cov(X,Y) = 25, Var(X) = 5, slope = 5.00;  d/dx E[Y | do(X=x)] = 3
simulation (n=2000, seed=123):  observational slope = 5.03
simulation of do(X=x) on 30 grid points, 5000 draws each (seed=7):
interventional slope = 3.00
bias formula a*c/(a^2+1) = 2*5/(4+1) = 2.00;  b + bias = 5.00
```

Worked Example 13.6 (A numeric counterfactual)**.** In the wage SCM of Example 13.1, suppose that for one specific individual we observe all three endogenous variables: ability $A = 1$ (for this illustration we assume a test measured it), education $X = 2.5$ and income $Y = 12.2$. *Counterfactual question*: what would this same person's income have been with 2 additional years of education, i.e. $X' = 2.5 + 2 = 4.5$, *holding their personal ability and personal luck (U_Y) fixed*?

1. *Abduction.* Each equation is additive in its noise, so we invert it: $U_A = A = 1$, $U_X = X - 2A = 2.5 - 2 = 0.5$, $U_Y = Y - 3X - 5A = 12.2 - 7.5 - 5 = -0.3$.
2. *Action.* Replace $X := 2A + U_X$ by $X := 4.5$; keep $A := U_A$ and $Y := 3X + 5A + U_Y$.
3. *Prediction.* $Y' = 3(4.5) + 5(1) + (-0.3) = 13.5 + 5 - 0.3 = 18.2$.

The counterfactual effect for this individual is $18.2 - 12.2 = 6.0$, exactly 3×2 additional years — because the structural coefficient on X is homogeneous (3) across individuals in this simple linear model (Theorem 13.5); in a model with individual-level effect heterogeneity (coefficients depending on A or U), different individuals could have different counterfactual gains, a theme central to Session 15's treatment-effect heterogeneity.

PYTHON LAB · s13_examples.py · ex_counterfactual_individual()

```
abduction:  U_X = X - 2A = 0.5,  U_Y = Y - 3X - 5A = 12.2 - 7.5 - 5 = -0.3
factual:  X = 2.5, Y = 12.2;  counterfactual X' = 4.5:  Y' = 18.2,
individual effect = 6.0
```

PART V

SCMs Across Artificial Intelligence

World models, agents that ask "what if", and credit among many agents

PART V · AI CONNECTIONS

13.11 SCMs in Agents, RL and Multi-Agent Systems

13.11.1 Agentic AI: what-if simulation and blame

An LLM agent deciding whether to call a costly tool is asking an interventional question about its environment, $P(\text{success} \mid \text{do}(\text{tool}=1))$ versus $\text{do}(\text{tool}=0)$; after the episode, a reflective agent (course session 26) asks a counterfactual one: *was the tool call necessary for this success?* That is exactly PN.

Worked Example 13.7 (Did calling the tool cause the success?)**.** From randomized A/B runs an agent platform knows $p_1 = P(S{=}1 \mid \text{do}(T{=}1)) = 0.80$ and $p_0 = P(S{=}1 \mid \text{do}(T{=}0)) = 0.30$ (illustrative numbers). By Proposition 13.2, $\max(0,\, 0.80 - 0.30) = 0.50 \le \text{PNS} \le \min(0.80,\, 1 - 0.30) = 0.70$. If the tool can never *hurt* (monotonicity) and tool use was randomized (exogeneity), $\text{PNS} = p_1 - p_0 = 0.50$, $\text{PN} = (p_1 - p_0)/p_1 = 0.5/0.8 = 0.6250$ and $\text{PS} = (p_1 - p_0)/(1 - p_0) = 0.5/0.7 = 0.7143$. Two SCMs with the same p_1, p_0 show the bounds are sharp (notation of the proof of Proposition 13.2: $p_1 = a + h$, $p_0 = a + k$, $\text{PNS} = h$): the monotone one has response types (always, helped, hurt, never) $= (0.30, 0.50, 0, 0.20)$, so $p_1 = 0.30 + 0.50 = 0.80$, $p_0 = 0.30 + 0 = 0.30$ and $\text{PNS} = 0.50$; the non-monotone one has $(0.10, 0.70, 0.20, 0)$, so $p_1 = 0.10 + 0.70 = 0.80$, $p_0 = 0.10 + 0.20 = 0.30$ and $\text{PNS} = 0.70$. Experiments alone cannot tell them apart; the monotonicity assumption is a substantive claim about the tool.

PYTHON LAB · s13_examples.py · ex_probabilities_causation()

```
PNS bounds:  [max(0, 0.80-0.30), min(0.80, 1-0.30)] = [0.50, 0.70]
under monotonicity:  PNS = 0.50,  PN = (p1-p0)/p1 = 0.6250,  PS =
(p1-p0)/(1-p0) = 0.7143
monotone SCM     : always=0.30 helped=0.50 hurt=0.00 never=0.20 -> p1=0.80,
p0=0.30, PNS=0.50
non-monotone SCM: always=0.10 helped=0.70 hurt=0.20 never=0.00 -> p1=0.80,
p0=0.30, PNS=0.70
```

13.11.2 Reinforcement learning: world models as SCMs

A model-based RL agent (course session 20) learns a transition model; written as $s_{t+1} := f(s_t, a_t, u_t)$, $r_t := g(s_t, a_t)$ with independent u_t, it is a Markovian SCM unrolled in time, and a policy is a soft intervention on the action nodes. It answers two different questions: the *interventional* expected cost of a new policy (fresh noise), and the *counterfactual* outcome of *this* logged episode under the new policy (same noise), the basis of counterfactual data augmentation. Planning (course sessions 23–24) is intervention on the same SCM: a STRIPS action schema is a structural assignment whose frame assumption ("everything else stays the same") is exactly the autonomy of the other mechanisms.

13.11.3 Multi-agent AI: counterfactual credit assignment

In cooperative multi-agent RL all agents share one team reward, so each agent's gradient is swamped by its teammates' choices. Write s for the global state, $\mathbf{u} = (u_1, \dots, u_N)$ for the joint action, $\mathbf{u}^{-i}$ for the actions of all agents except i, τ^i for agent i's action–observation history, $\pi^i(u_i \mid \tau^i)$ for its policy, and $Q(s, \mathbf{u})$ for a centralized critic's estimate of the team return. COMA (Foerster et al. 2018) gives agent i the counterfactual advantage (the superscript i distinguishes it from the ability variable A of the wage SCM)

$$A^i(s, \mathbf{u}) = Q(s, \mathbf{u}) - \sum_{u_i'} \pi^i(u_i' \mid \tau^i)\, Q\big(s, (\mathbf{u}^{-i}, u_i')\big),$$

"how much better was my action than what I would have done on average, *holding my teammates' actions fixed*?" The baseline is a rung-3 quantity: it intervenes on agent i's action node while the other agents' realized actions are kept. Because the baseline b (the sum above) does not depend on u_i, it leaves the policy gradient unbiased: $\mathbb{E}_{u_i \sim \pi^i}[\nabla \log \pi^i(u_i)\, b] = \sum_{u_i} \pi^i(u_i) \frac{\nabla \pi^i(u_i)}{\pi^i(u_i)}\, b = b \nabla \sum_{u_i} \pi^i(u_i) = b \nabla 1 = 0$.

Worked Example 13.8 (COMA on a two-agent matrix)**.** Two agents with actions $\{a, b, c\}$ have centralized critic values $Q(u_1, u_2)$ (state fixed, illustrative):

$$Q = \begin{array}{c|ccc} & u_2 = a & u_2 = b & u_2 = c \\ \hline u_1 = a & 2 & 0 & 1 \\ u_1 = b & 4 & 3 & 0 \\ u_1 = c & 1 & 5 & 2 \end{array}$$

Policies over (a, b, c) are $\pi^1 = (0.5, 0.3, 0.2)$ and $\pi^2 = (0.4, 0.4, 0.2)$, and the joint action played was $(u_1, u_2) = (b, a)$ with $Q(b, a) = 4$. Agent 1's baseline varies its own action with agent 2 fixed at a (column a): $0.5 \cdot Q(a, a) + 0.3 \cdot Q(b, a) + 0.2 \cdot Q(c, a) = 0.5(2) + 0.3(4) + 0.2(1) = 1.0 + 1.2 + 0.2 = 2.4$, so $A^1 = 4 - 2.4 = 1.6$. Agent 2's baseline varies its action with agent 1 fixed at b (row b): $0.4(4) + 0.4(3) + 0.2(0) = 1.6 + 1.2 + 0 = 2.8$, so $A^2 = 4 - 2.8 = 1.2$. Both receive positive but different credit from the same team reward.

PYTHON LAB · `s13_examples.py` · `ex_coma_baseline()`

```
agent 1:  baseline = 2.40, advantage = 1.60;  agent 2:  baseline = 2.80,
advantage = 1.20
```

PART V · AI CONNECTIONS

13.12 Summary

Table 13.1. Cheat-sheet of key formulas.

Object	Formula
SCM	$V_i := f_i(\mathrm{pa}_i, U_i)$, $P(\mathbf{U})$; Markovian if U_i independent
Causal Markov	$P(\mathbf{v}) = \prod_i P(v_i \mid \mathrm{pa}_i)$, $P(v_i \mid \mathrm{pa}_i) = P(f_i(\mathrm{pa}_i, U_i) = v_i)$
Global Markov	$\mathbf{X} \perp\!\!\!\perp_d \mathbf{Y} \mid \mathbf{Z}$ in $G \Rightarrow \mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}$ in P
Truncated factorization	$P(\mathbf{v} \mid \mathrm{do}(x)) = \prod_{i:V_i \neq X} P(v_i \mid \mathrm{pa}_i)\big\vert_{X=x}$
Counterfactual	abduction $P(\mathbf{u} \mid e) \to$ action $M_{x'} \to$ prediction $Y_{x'}(\mathbf{u})$
PNS bounds	$\max(0, p_1 - p_0) \le \mathrm{PNS} \le \min(p_1, 1 - p_0)$; monotone: $p_1 - p_0$
Linear SEM	$\Sigma = (I - B)^{-1}\Omega(I - B)^{-\top}$
Omitted-variable bias	$\boldsymbol{\beta} = \mathbf{b} + \Sigma_{XX}^{-1}\Sigma_{XA}\mathbf{c}$; scalar: $b + ac\sigma_A^2/(a^2\sigma_A^2 + \sigma_X^2)$
COMA advantage	$Q(s, \mathbf{u}) - \sum_{u_i'} \pi^i(u_i')Q(s, (\mathbf{u}^{-i}, u_i'))$

THINK ABOUT IT **The causal Markov condition needs independent noises. Which unmeasured factors would make the noises of "education" and "income" dependent, and how would you show that in the graph?**

THINK ABOUT IT **A world model in model-based RL is trained on logs from an old policy. Under which conditions is it structural (valid under the new policy) rather than merely predictive?**

THINK ABOUT IT **Why can randomized experiments identify PNS only up to an interval? What extra knowledge about an LLM agent's tool would collapse the interval?**

Further reading. Pearl (2009), *Causality*, chs. 1, 3, 7, 9; Pearl, Glymour and Jewell (2016), *Causal Inference in Statistics: A Primer*, chs. 2–4; Peters, Janzing and Schölkopf (2017), *Elements of Causal Inference*, chs. 3–6; Spirtes, Glymour and Scheines (2000), *Causation, Prediction, and Search*; Hernán and Robins (2020), *Causal Inference: What If*, ch. 6; Koller and Friedman (2009), ch. 21; Foerster et al. (2018), "Counterfactual multi-agent policy gradients", AAAI.

CONNECTIONS AND TAKEAWAYS

We now have a mathematical object — the SCM — in which "the effect of an intervention" and even "what would have happened to this specific unit" are well-defined quantities, computed by the truncated factorization formula (Theorem 13.3) directly from the same graphical and computational machinery built in Sessions 6–7. The causal Markov condition (Theorem 13.1) tells us exactly which observable constraints a causal model imposes, and d-separation reads them off the graph. This resolves Session 12's conceptual gap in principle.

But notice the catch in every worked example above: we always assumed we had the *full structural equations*, including the confounder A and its exact numerical role. In practice we almost never observe every relevant confounder, let alone know its functional form. **Session 14** asks the sharper, practitioner-relevant question: if all we are given is an observational dataset plus a causal *graph* (no equations, and possibly some variables, like A, unmeasured, i.e. bidirected edges) can we still compute $P(y \mid \mathrm{do}(x))$ using only the pieces we can actually observe? The answer will be: sometimes yes, via a checkable graphical criterion — the back-door and front-door criteria and the do-calculus — and sometimes provably no.